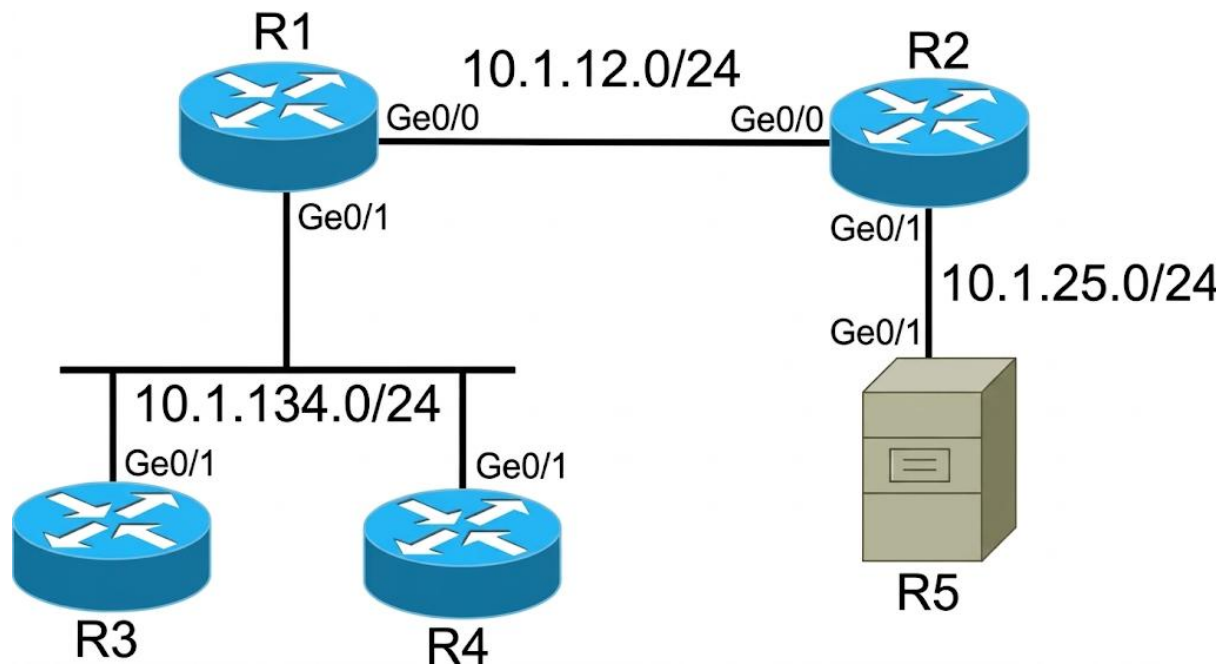


Static NAT



Objective

Configure and verify Static Network Address Translation (Static NAT) to map specific private internal IP addresses to dedicated public IP addresses on a one-to-one basis.

Tasks

- Configure hostnames and IP addresses on all network devices.
- Disable routing on **R3** and **R4**, configuring their default gateway as **10.1.134.1**.
- Disable routing on **R5** and configure its default gateway as **10.1.25.2**.
- Configure a default route on **R1** pointing to **10.1.12.2** as the next hop.
- Enable HTTP service on **R5**.
- Configure Static NAT mappings on **R1**:
 - **10.1.134.3** => **10.1.12.3**
 - **10.1.134.4** => **10.1.12.4**
- Verify end-to-end connectivity from **R3** and **R4** to **R5**.

Concept & Overview

Network Address Translation is a fundamental boundary technology implemented in enterprise and service provider networks. NAT translates RFC 1918 private IP address space into globally routable public IP addresses. It conserves public IPv4 addresses and enhances security by masking internal topology from external entities.

Static NAT performs explicit one-to-one mapping, requiring an equal number of private and public addresses. In this topology, the **10.1.12.0/24** network simulates public address space between **R1** and **R2**. Before defining translation pairs, inside and outside interfaces must be designated so the router correctly identifies boundary transit.

Device Configurations

Router R1

```
enable
```

```
configure terminal
```

```
hostname R1
```

```
!
```

```
interface GigabitEthernet0/1
```

```
ip address 10.1.134.1 255.255.255.0
```

```
ip nat inside
```

```
!
```

```
interface GigabitEthernet0/0
```

```
ip address 10.1.12.1 255.255.255.0
```

```
ip nat outside
```

```
!
```

```
ip route 0.0.0.0 0.0.0.0 10.1.12.2
```

```
!
```

```
ip nat inside source static 10.1.134.3 10.1.12.3
```

```
ip nat inside source static 10.1.134.4 10.1.12.4
```

```
!
```

```
end
```

Router R2

```
enable
```

```
configure terminal
```

```
hostname R2
```

```
!
```

```
interface GigabitEthernet0/1
```

```
ip address 10.1.25.2 255.255.255.0
```

```
no shutdown
```

```
!
```

```
interface GigabitEthernet0/0
```

```
ip address 10.1.12.2 255.255.255.0
```

```
no shutdown
```

```
!
```

```
end
```

Router R3

enable

configure terminal

hostname R3

!

interface GigabitEthernet0/1

ip address 10.1.134.3 255.255.255.0

no shutdown

!

no ip routing

!

ip default-gateway 10.1.134.1

!

end

Router R4

enable

configure terminal

hostname R4

!

interface GigabitEthernet0/1

ip address 10.1.134.4 255.255.255.0

```
no shutdown
```

```
!
```

```
no ip routing
```

```
!
```

```
ip default-gateway 10.1.134.1
```

```
!
```

```
end
```

Router R5

```
enable
```

```
configure terminal
```

```
hostname R5
```

```
!
```

```
interface GigabitEthernet0/1
```

```
ip address 10.1.25.5 255.255.255.0
```

```
no shutdown
```

```
!
```

```
no ip routing
```

```
!
```

```
ip default-gateway 10.1.25.2
```

```
!
```

end